

Q2) Write your observation and conclusions for the following:

- A few marbles are put in a glass half filled with water
- Ice kept at room temperature

Ans a) The water level will increase and it proves that marbles have mass and they occupy space

b) Ice will melt

SAT

1) Give 2 examples for the following

- liquid milk water
- Dry Ice, camphor
- Ink, oxygen gas

II Key notes

* Define the following

Matter - Matter is anything that has mass occupies space and can be perceived by our senses

Atom - Atom is the smallest possible unit of matter that exhibits all properties of matter

Evaporation - The process by which a substance in its liquid state changes to its vapour state is called evaporation

Sublimation - The process by which a substance directly changes from its solid state to its gaseous state without passing through liquid is called sublimation

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III

Answer the following

- How can you show that particles of matter are in continuous random motion?

Ans Take a bowl with half filled water in it. Sprinkle a pinch of saw dust on its surface.

When observed case fully these particles will be in continuous movement. This shows that particles of matter are in

random motion

- 2) Why do the particles of matter move?
How does this movement change with an increase in temperature?

ANS Due to Kinetic energy of particles movement of particles increases with increase in temperature.

- 3) Differentiate between an atom and a molecule

Atom	Molecule
Smallest particle of matter that does not exist independently.	Smallest part of matter that exists independently.

- 4) Why do solids, liquids and gases differ in their physical state?

Ans Arrangement of particles in all three states of matter differ from each other due to difference in intermolecular attraction or intermolecular space.

Give Reasons

- 1) When a stone is dipped in a glass containing some water, the level of water rises, but ~~when~~ when a ~~few~~ spoon of sugar is added to it and stirred, the water level does not rise.

ANS Stone is an insoluble solid when dropped in water, a rise in the level of water is seen.

Sugar particles are soluble in water and occupy intermolecular space, therefore the level of water remains same.

- 2) Naphthalene balls disappear with time without leaving any solid behind.

ANS Naphthalene balls sublime and solid gets converted to vapour directly changing to gaseous state.

- 3) All gases are fluids but all fluids are not gases.

ANS Particles with large intermolecular space and low force of attraction only can follow.

Physical and chemical changes

T ~~Keywords~~ KW)

K W L

T Keywords

- 1) Periodic change
- 2) Desirable change
- 3) Reversible change
- 4) Sublimate
- 5) ~~Condensation~~
- 6) Dissolution

II Define the following terms

- 1) Natural change - A change in Nature that takes place on its own and continues forever and can generally not be influenced by human effort.
- 2) Desirable change - A change that is useful for human kind or nature
- 3) Burning - A chemical change in which a substance reacts with oxygen or air to produce an oxide with an evolution of energy in the form of heat and light.
- 4) Reversible change - change that can be reversed by changing the external conditions, they are mostly temporary.

III Answer the following questions

- 1) Human made changes have impacted ^{Nature} positively and negatively. Discuss and give 2 examples to support.

2) How can you say that when zinc is added to dilute hydrochloric acid a chemical change takes place?

ANS When zinc is added to dilute hydrochloric acid bubbles are observed in the test tube indicating the evolution of gas which in turn proves that a chemical reaction ^{has} taken place.

3) How can we check the evolution of Hydrogen gas in a chemical ~~change~~ ^{reaction}

ANS If a lighted matchstick is brought near the mouth of a test tube in which a chemical reaction has taken place and Hydrogen gas is evolved then a pop sound heard and the flame of the match stick gets extinguished.

4) A burning candle shows both physical and chemical changes explain

1) Mention a change which ^{is} always
a) desirable b) undesirable c) periodic

ANS a) Desirable - Ripening of fruits

b) Undesirable - Rusting of Iron

c) Periodic change - Changing of seasons

2) Name a chemical change which takes place in
presence of a) heat b) light c) electricity

ANS a) Heating of Calcium Carbonate

b) Photosynthesis in plants

c) Electrolysis of water into hydrogen and oxygen.

4) ANS When a candle is burned the wax melts and solidifies. This shows a physical change. When twig is burning and produces a flame it releases a flame, water vapor, carbon dioxide gas and smoke signifying a chemical change.

IV Differentiate between

1) Evaporation & boiling

Evaporation

- It is a slow process
- It takes place
- It happens at all temperature

Boiling

- It is a fast process
- It takes place at all parts of the liquid
- It ~~is~~^{happens} at fixed temperature

2) Physical change

★ temporary change

★ can be reversed

No new substances are formed

★ same chemical composition

Chemical change

★ Permanent change

★ can not be reversed

★ New substances

★ Composition, properties, change

V Give reason

ripening of fruits is a chemical change

During the ripening of fruits a new product is formed with different properties such as taste, texture and smell.

2) Fermented Food is warmer than ordinary Food

ANS During fermentation of food, chemical reactions take place, this leads to evolution of carbon dioxide and heat.

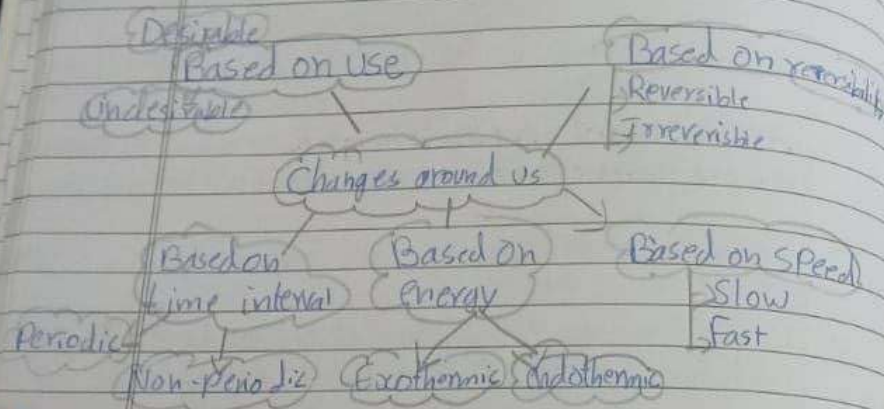
4) Wet clothes dry faster during summer compared to rainy or winter days

During summers the temperature is high while humidity is low as compared to winter or rainy days.

3) Burning is a chemical change

Chemical change is observed while burning a paper. New substances as oxide of elements, carbon dioxide, heat, light are formed and evolved.

Concept Map-1



Concept map-2

Changes

Physical

Chemical

Temporary

Permanent

No new substance formed

always a new substance

Reversible

Irreversible

heat or light may or may not be given out or consumed

heat or light is a part of chemical change may be given out or consumed

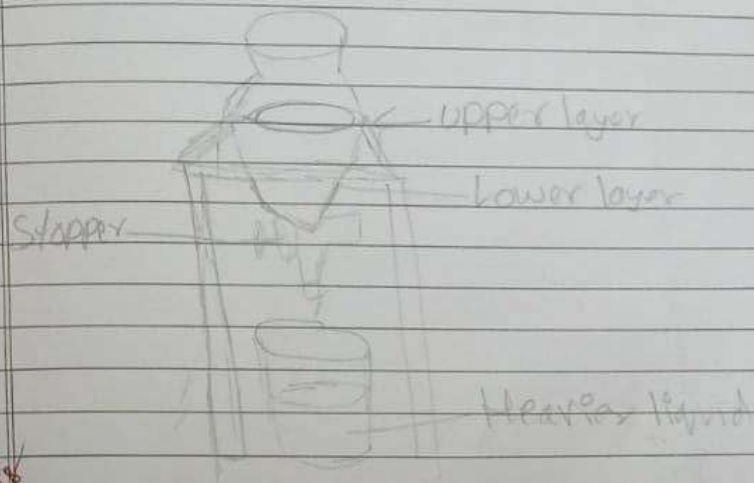
Original substance can be obtained

Original substance can be obtained by physical method

HW

6 Draw a well labelled diagram to show separation of a mixture of kerosene oil and water

Ans



Differentiate between
compounds

Mixture

- ★ 2 or more elements chemically combined in a fixed ratio by mass.
- ★ a compound has a definite chemical formula.
- ★ It is always homogeneous.
- ★ It cannot be separated by physical methods.
- ★ Show new set of properties compared to its constituent elements.
- ★ Energy is either liberated or absorbed during formation of a compound.
- ★ 2 or more elements or compounds are not combined chemically in a fixed ratio by mass.
- ★ A mixture does not have a fixed or definite chemical formula.
- ★ It can be heterogeneous.
- ★ It can be separated by physical methods.
- ★ Individual properties of the constituent components are retained.
- ★ There is no necessary liberation or absorption of energy during the

Distillation

Fractional Distillation

- ★ It can be used for ~~liquid-solid~~ liquid-liquid mixtures.
- ★ It is used for separation of liquids with large difference in boiling points.
- ★ Only a single process of evaporation takes place (Condensation cycle is used).
- ★ It is used for liquid-liquid mixture only.
- ★ It can be used for separation of liquids with small difference in boiling point.
- ★ Here multiple vaporization process takes place (Repeated condensation cycle is used).

Give reason for the following

(Q1) All elements and compounds are pure substances

Ans Elements and compounds have definite composition with a definite set of properties they are always homogeneous in nature.

(Q2) The mixture of 2 or more gases is always a homogeneous mixture

ANS All gases mix with each other in a uniform proportion and gives an appearance of homogeneous mixture

Q3) Milk is a mixture and is not a pure substance

ANS Milk is made up of Fats, carbohydrates, Proteins, Vitamins, Salts and Water. The proportions of all these constituents may vary from milk to another.

Q4) Salt can be separated from Ammonium chloride using sublimation but a mixture of salt and sand cannot be separated by same method